

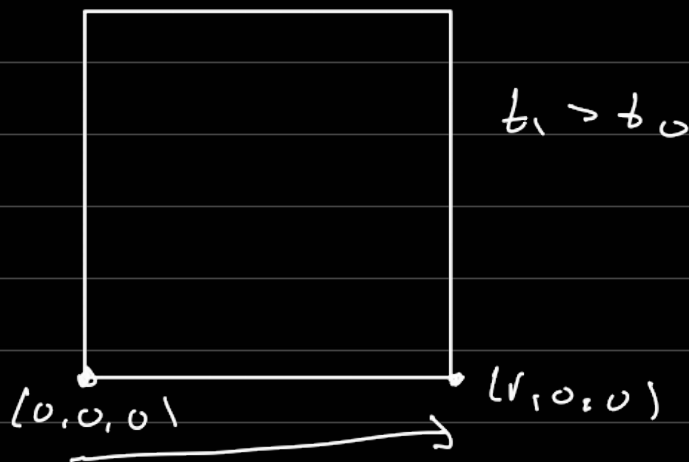
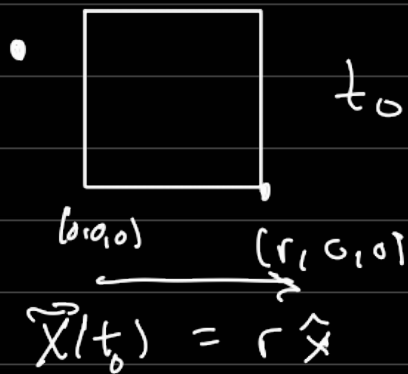
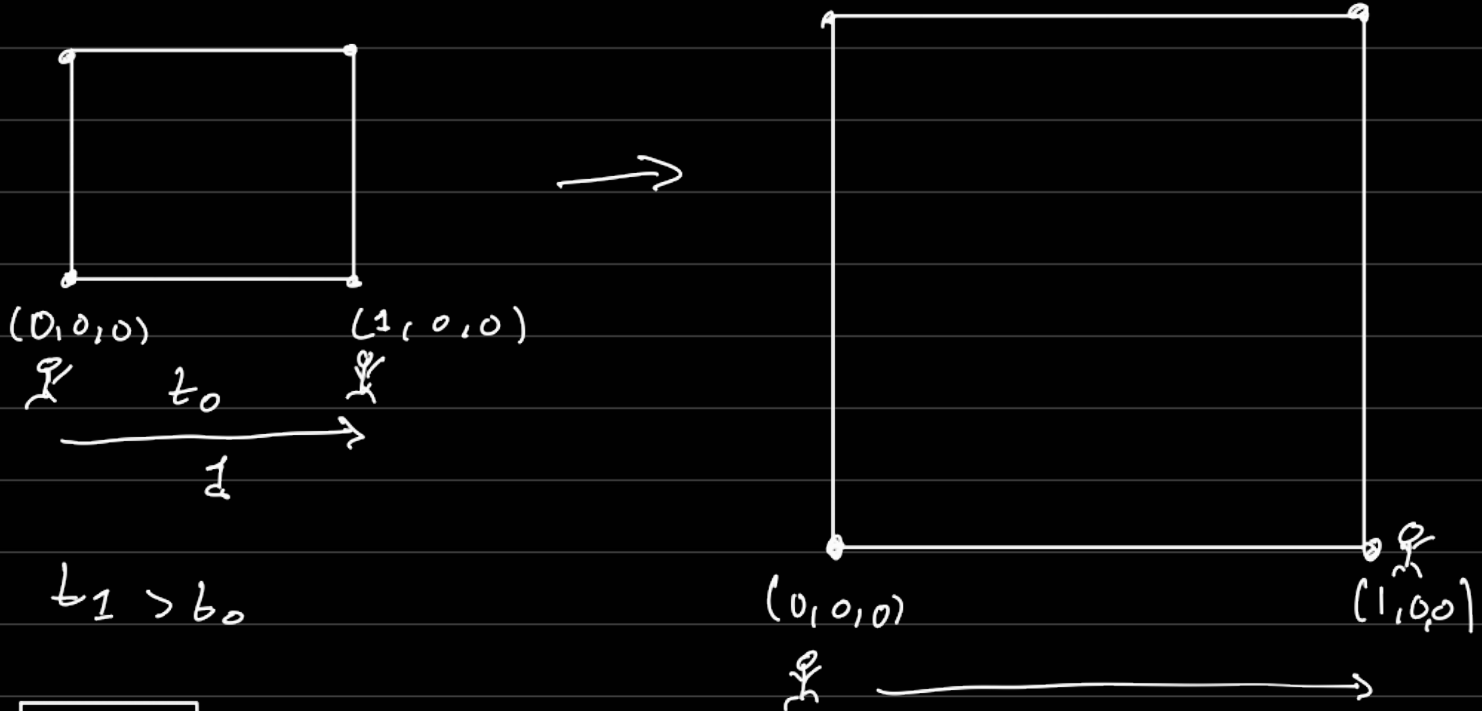
Cosmology:

* Largest scales: galaxy clusters $\rightarrow$ cosmic web.

* "Fundamental observers": $v_{pec} = 0$ *

$$v_{obs} = \underbrace{H_0 D}_{H_{Hubble}} + v_{pec}$$

$\rightarrow$ How to interpret the expansion:



$R(t)$: scale factor
(cosmological)

$$\vec{X}(t) = R(t) r \hat{x}$$

cosmological $|r| = \frac{|\vec{X}|}{R(t)}$

Recall: $V = HD$ $12(t_0)$

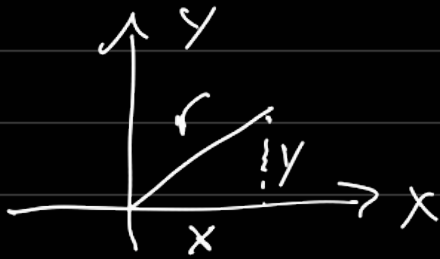
$$\frac{d}{dt} \vec{X}(t) = \vec{V}(t) = \dot{R}(t) \cdot \hat{x} = \frac{\dot{R}(t)}{R(t)} \vec{X}(t)$$

$$\vec{V}(t) = \left(\frac{\dot{R}}{R} \right) \vec{X}(t_0)$$

$\underbrace{\hspace{1cm}}_{\sim H} \quad \underbrace{\hspace{1cm}}_D$

* GR: "metric": measuring tool

2D:



$$r^2 = y^2 + x^2$$

ds^2 : ds : "infinitesimal distance": line element



$$ds^2 = dx^2 + dy^2 \quad *$$

$$(x, y) \rightarrow (r, \theta)$$

* $\rightarrow$ Coordinates: $x = r \cos(\theta)$, $y = r \sin(\theta)$

o prod: $dx = dr \cos(\theta) - r \sin(\theta) d\theta$

$$dy = dr \sin(\theta) + r \cos(\theta) d\theta$$

$$dx^2 + dy^2 = ds^2$$

* $ds^2 = dr^2 + r^2 d\theta^2 \quad *$ (Metric)

* Metric: can measure / understand lengths & time in more complex systems

Time: Metric is invariant: space + time



$$D^2 = x^2 + y^2 + z^2$$

$$(c\Delta t)^2 = D^2 = x^2 + y^2 + z^2$$

$$(c\Delta t)^2 - x^2 - y^2 - z^2 = 0 \quad x \rightarrow dx \text{ etc.}$$

$$\rightarrow \text{smaller } (c\Delta t)^2 - \underbrace{dx^2 + dy^2 + dz^2}_{\substack{\uparrow \\ \text{measured by a person}}} = 0 \rightarrow \text{also}$$

$$\boxed{(c\Delta t)^2 - dx^2 - dy^2 - dz^2 = 0} \quad (\text{invariant - line interval})$$

$$(ds^2)_{\text{light}} = 0 = (cdt)^2 - dx^2 - dy^2 - dz^2$$

* look at massive particles ...

$$\boxed{(ds^2) = (cdt)^2 - dx^2 - dy^2 - dz^2}$$

metric for Minkowski space-time in S.R.

* Friedmann model:

$$\boxed{H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2}}$$

* Critical density: $k=0$, t

$$\bullet \rho_c = \frac{H^2 3}{8\pi G} \quad | \quad \Omega(t) = \frac{\rho(t)}{\rho_c} \quad \left(\begin{array}{l} \text{dim-less} \\ \text{density parameter} \end{array} \right)$$

Plug $\rho(t) = \rho_c \Omega(t)$ in F.E.

$$\bullet \Omega(t) = 1 + \frac{kc^2}{\dot{a}^2(t)}$$

3 different types of models:

(i) Matter ρ_m

(ii) Radiation ρ_r

(iii) Vacuum ρ_Λ

$$\rightarrow \text{F.E.: } \underbrace{G_{\mu\nu} + \Lambda g_{\mu\nu}}_{\substack{\text{grav.} + \text{Vacuum}}} = \underbrace{T_{\mu\nu}}_{\substack{\uparrow \\ \text{matter}}}$$

$\rightarrow$ Einstein's field equations

$$P = \omega \rho c^2$$

ω : varies for all i-iii

(i) Matter $\omega = 0$: $mc^2 \gg K.E.$

$\rightarrow$ Low K.E. low temp $\Rightarrow$ low pressure

(ii) Radiation: $\omega = 1/3$
 $\rightarrow$ S.M. $P \approx \frac{1}{3} \epsilon \sim \frac{1}{3} n_\gamma E_\gamma$ ← energy density

(iii) Vacuum $\omega = -1$

Model building: A.6

i) Matter:

$$\rho \sim \frac{M}{Vol}$$

as expands $V \sim \frac{4\pi}{3} |\vec{x}|^3 = \frac{4\pi}{3} r^3 R^3(t)$

⑤ Matter conservation: M is constant

$$\rho(t) \sim \frac{1}{R(t)^3} \Rightarrow \rho_m(t) = \rho_{0,m} \left(\frac{R_0}{R(t)} \right)^3$$

• Aside: Any term that has 0

cr: ρ_0, Ω_0, R_0

Refers to the values as they today

ii) Radiation: Energy densities

$$\epsilon_r = \rho_r = n_\gamma E_\gamma$$

$$n_\gamma \sim \frac{N}{Vol} \sim \frac{N}{\frac{4\pi}{3} R^3} \sim \frac{1}{R^3}$$

$$E_\gamma = h\nu = \frac{hc}{\lambda} = \frac{hc R_0}{R(t) \lambda_0} \sim \frac{1}{R(t)}$$

$$\frac{\lambda_0}{R_0} = \frac{\lambda}{R(t)} \Rightarrow \lambda = R(t) \frac{\lambda_0}{R_0}$$

$$\rho_r \sim \frac{1}{R^4} \Rightarrow \rho_r(t) = \rho_{0,r} \left(\frac{R_0}{R(t)} \right)^4$$

list) Vacuum: core back

list) Matter: $\rho_m(t) = \rho_{0,m} \left(\frac{R_0}{R}\right)^3$
Want to know how $R(t)$ looks

$$\left(\frac{\dot{R}}{R}\right)^2 = \frac{8\pi G}{3} \rho_m - \frac{Kc^2}{R^2}$$

Plug in $= \frac{8\pi G}{3} \left(\rho_{0,m} \left(\frac{R_0}{R}\right)^3 \right) - \frac{Kc^2}{R^2}$

$$\dot{R}^2 = \frac{8\pi G}{3} \rho_{0,m} \frac{R_0^3}{R} - Kc^2$$

$$K = 0, 1, -1$$

↑ ↑ ↑

flat, spherical, Hyperbolic
(Closed) (Open)

$K=0$: flat: EdS - space time

• Matter filled, flat space time

$$\dot{R} = \frac{8\pi G}{3} \rho_{0,m} \frac{R_0^3}{R} \quad | R_*$$

solve this:

$$\frac{dR}{dt} = \alpha \frac{1}{R}$$

$$\int_0^R R dR = \int_0^t \alpha dt$$

$$\frac{1}{2} R^2 = \alpha t \Rightarrow R = \frac{\sqrt{\alpha}}{\sqrt{2}} t^{1/2}$$

$$R(t) \sim t^{1/2}$$

B.3 Exam:

$$z = \frac{\lambda_{obs} - \lambda_{em}}{\lambda_{em}} \sim \frac{\Delta \lambda}{\lambda_{em}}$$

$$z = \frac{\lambda_{obs}}{\lambda_{em}} - 1 = \frac{\lambda_{obs}}{\lambda_{em}} - 1$$

① $\boxed{z+1 = \lambda_{obs}/\lambda_{em}}$ $\left| \quad \cancel{\frac{\lambda_0}{R_0}} = \frac{\lambda(t)}{R(t)} \quad \right|$

$\boxed{\frac{R_0}{R(t)} = \frac{\lambda_0}{\lambda(t)} = z+1} \quad *$

$\rho_m(t) = \rho_{m,0} \left(\frac{R_0}{R(t)} \right)^3 = \rho_{m,0} (z+1)^3$

$H^2 = \left(\frac{\dot{R}}{R} \right)^2 = \frac{8\pi G}{3} \rho_m - \frac{Kc^2}{R^2} \rightarrow 0$

for EdS $K=0$

$H^2 = \frac{8\pi G}{3} \rho_{m,0} (z+1)^3$

$\boxed{H = \sqrt{\frac{8\pi G \rho_{m,0}}{3}} (z+1)^{3/2}} \quad *$

$H_0 = \sqrt{\frac{8\pi G \rho_{m,0}}{3}}$

② also Q4 (a)

$ds^2 = (cdt)^2 - R(t) (dr^2 + S_k(r) d\Omega^2) \quad *$

$K=0, d\theta, d\phi = 0$

$\underline{ds^2} = (cdt)^2 - R^2(t) dr^2 = 0 \quad *$

$(cdt)^2 = R^2(t) dr^2$

$\boxed{dr = \pm \frac{cdt}{R(t)}}$

$+$: light moving away
 $-$: light moving towards us

Int. $\int dr = \int_0^t \frac{cdt}{R(t)}$

$r_{ph}(t) = c \int_0^t \frac{dt}{R(t)}$

mult. by R_0 :

$\boxed{R_0 r_{ph}(t) = c \int_0^t \left(\frac{R_0}{R(t)} \right) dt}$

$$H^2 = \left(\frac{\dot{R}}{R}\right)^2$$

$$\frac{R_0}{R} = 1+z$$

d.f.f: $\frac{d}{dt} \left(\frac{R_0}{R} \right) = \frac{d}{dt} (1+z)$

$$-\frac{R_0}{R^2} \dot{R} = \frac{dz}{dt}$$

$$dz = dt \left(-\frac{R_0}{R} \frac{\dot{R}}{R} \right)$$

$\underbrace{\quad}_{H}$

$$t=0:$$

$$-dz/H = dz \frac{R_0}{R} \quad \left| \begin{array}{l} \frac{R_0}{R} = 1+z \\ R(0) \rightarrow 0 \end{array} \right.$$

$$R_0 r_{ph}(z) = c \int_{\infty}^z \left(-\frac{dz}{H(z)} \right)$$

$$t \rightarrow 0, z \rightarrow \infty$$

$$t > 0, z$$

$$R_0 r_{ph}(z) = c \int_z^{\infty} \frac{dz}{H(z)}$$

start of
Q4 B + EB.3
(b)

→ finish EB.3 (b) EdS

$$R_0 r_{ph}(z_{rec}) = c \int_{z_{rec}}^{\infty} \frac{dz}{H_0 (1+z)^{3/2}} \quad \leftarrow \text{from (c)}$$

$$= \frac{c(-z)}{H_0} (1+z)^{-1/2} \Big|_{z_{rec}}^{\infty}$$

$$= 0 - \left(\frac{c(-z)}{H_0} (1+z_{rec})^{-1/2} \right)$$

$$R_0 r_{ph}(z_{rec}) = \frac{zc}{H_0} (1+z_{rec})^{-1/2}$$

